

Project Update Report: FAQ & Q&A; Platform

1. Complete UI/UX Modernization (Glassmorphism & CyberShield Theme)

- Premium Aesthetics: Transitioned the platform from a standard Bootstrap layout to a modern 'SaaS' aesthetic featuring a Deep Slate dark mode.
- Glassmorphism UI: Implemented dynamic glowing radial background gradients and applied heavy backdrop filters to create a highly premium glassmorphic effect across all components.
- Modern Typography: Replaced default browser fonts with sleek, modern web fonts (Outfit and Inter).

2. Reddit-Style Threaded Q&A; Architecture

- Nested Tree Structure: Upgraded the simple Q&A; layout to a full Reddit-style threaded tree. Comments and replies can be infinitely nested.
- Interactive Threadlines: Implemented visual vertical threadlines that connect parent comments to their children. These threadlines dynamically illuminate on hover.
- Minimalist Action Bars: Replaced bulky UI buttons with classic horizontal action bars featuring inline upvote/downvote scores, reply buttons, and compressed inline avatar headers.

3. High-Concurrency Database Optimizations

- Scalability for 2000-3000 Users: Anticipating heavy concurrent usage, optimized the underlying database architecture to prevent performance bottlenecks.
- N+1 Query Resolution: Heavily utilized Django's `select_related` and `prefetch_related` ORM methods in the views to ensure the complex nested comment trees can be queried and rendered in a single, highly efficient database trip.

4. YakshaBot AI Auto-Answer Integration (RAG)

- Instant AI Responses: Bridged the static FAQ database with the live Community Q&A.; The moment a user posts a new question, the backend instantly runs a Retrieval-Augmented Generation (RAG) query against published FAQs.
- Auto-Posting & Citations: If relevant context is found, a system profile named 'YakshaBot' automatically generates and posts the first answer to the question instantly, complete with hyperlinks citing the exact FAQ sources used.
- Distinct AI Badging: Added custom robot avatars and 'AI' UI badges so users clearly recognize the response is automated.